

Ecolab

6–8 minutes

Since its inception in the Fall of 2014, the Ecolab has focused on chemical water quality analysis of the fluvial system in Ecolab. Recently, the Ecolab has expanded to include a variety of projects such as GIS, restoration work, and more. Ecolab is quickly becoming a valuable program and asset for the Southwestern University community as it allows for students from a variety of academic interests and backgrounds to conduct environmental field work in close proximity to campus. In this, students can easily apply skills and knowledge gained in the classroom to field work of their choosing. The following research themes are currently in progress:

Water Chemistry

Since August 2014, Ecolab has performed water quality analysis on samples collected following a significant rain event. Following each rain event, water samples are collected from four different locations in Ecolab; the cattail pond, stream, deer crossing, and culvert. Current water quality analyses include pH, turbidity, conductivity, chloride, nitrate, on-site pH, on-site dissolved oxygen, and phosphorus tests.

Fauna

An ongoing project, Ecolab is developing a census of animals through the use of GameCams. These cameras were obtained following a grant awarded from the King Creativity Fund. Thus far, the cameras have captured a variety of animals including: deer, raccoons, armadillo, coyote, rabbit, and bobcat. Numerous other signs of wildlife such as prints and snake-skins have been observed. Students continue to work toward developing this census along with analyzing patterns of animal movement throughout Ecolab.

Flora

Ecolab is working to develop a biannual vegetation survey that will identify and archive plants found within Ecolab. Students will also categorize them as invasive or noninvasive to allow for removal of invasive species.

Environmental History

Using Geographic Information Systems (GIS), students utilized historic aerial photography to show land use change from 1941 to present day. With this imagery, students showed how the fluvial system had drastically changed with the addition of the cattail pond and how the land use has shifted throughout time.

[Follow this link to see the Map](#)

GIS

In partnership with the GIS Labs, the Ecolab is has multiple projects underway including, a historic analysis the land through aerial photography, a fluvial model of the stream and pond system, photosphere technology and create a virtual tour of EcoLab.

Restoration

The overall goal for restoration of the Ecolab is to transform it into a workable classroom space while maintaining a workable space while maintain its research and pedagogical service. Restoration plans include the continued removal of trash, trail creation and maintenance, and promoting and protecting the area as an ecological site. Additionally, restoration work will include the cataloging and if possible removal of invasive species, and a plan to contain and curb cattail growth. We will be replacing the invasive species with native species. In terms of modifications, the goal of turning the area into a workable space will include a few updates. A primary goal of restoration work is to transform Ecolab into a workable classroom space. This will be accomplished by clearing and creating an area large enough to accommodate a class without any negative impacts on the ecosystem.

COURSES

Ecolab Practicum

ENV49-301-01&02

The purpose of this course is to expose students to applications of Environmental and Conservation Studies through (1)

environmental monitoring and (2) conservation/restoration projects.

Students will participate in environmental monitoring - collecting, analyzing and sharing information - from the Southwestern Ecolab. In this course environmental monitoring encompasses a variety of activities, including rigorous research and long-term surveillance. This course examines management and policy issues relating to conservation, while providing scientific background to understand these issues. Students will organize in a semester long project focusing on incremental progress in management of the Southwestern Ecolab.

Students are expected to log hours of work in the Ecolab (3 hours per week), maintain a course blog outlining experiences, summarize assigned readings' applications to Ecolab, and meet biweekly to discuss progress and concepts.

Ecolab II: Research

ENV49-301-03&04

Prerequisites: EcoLab I

This course is an intensive research practicum in a collaborative setting with faculty, fellow classmates, and introductory level students of Ecolab I. This course draws from experiences in Ecolab I and other coursework. Approach to study is expected to be integrative, drawing from multiple fields and past work done in the Ecolab. Students are expected to develop research questions, perform literature reviews, test hypotheses, and draw conclusions. At instructor's discretion, it maybe required to have an external faculty advisor for projects. Students are expected to be leaders in the Ecolab, present work to the Southwestern community, write a

scholar report, and meet biweekly to discuss progress and concepts.

PEOPLE

Faculty

- Romi Burks, *Professor of Biology and Environmental Studies*
- Laura Hobgood, *Professor of Religion and Environmental Studies*
- Joshua Long, *Assistant Professor of Environmental Studies*
- John Ore, *Professor of Theatre*
- Willis Weigand, *Director of General Chemistry Labs*

Current Students

- Simone Yoxall
- Keara Hudler
- Zane McDurham

Alumni of EcoLab

- Lauren Childres
- Dakota McDurham
- Nick Espino
- Caitlin Schneider
- Daniel Buffington
- Hailey Johnson